An advanced volcanic hazard evaluation by understanding the atmospheric effect on a surface-temperature: An application at Santorini volcano monitoring station, South Aegean Active Volcanic Arc.

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Abstract

The main aim of this research is to improve monitoring of an active volcano by understanding the multiple time scale components and variables that contribute to recorded measurements. We examine how the atmospheric air and the meteorological conditions that take place on Santorini volcano monitoring station affect the surface volcanic temperature recordings of the volcano. For this purpose, we use the time series analysis by the application of both the Multiple Linear Regression (MLR) and the Time Series Regression (TSR) models. The TSR model permits both the mathematical and physical interpretation of the utilized variables with the implementation of the lag operators. The time series decomposition has been applied into two time-scale components. The seasonal-term and the short-term components show a removal of the observed noise in the time series avoiding any possible misinterpretations. The total explanation of the predicted model is up to 70%, whereas the unexplained remnant part displays that endogenic reactions and surface deformation that may also influence the volcanic temperature of the Santorini volcano. This methodology can be applied to other volcanic monitoring systems to improve volcanic monitoring and provide an advanced volcanic hazard evaluation.

Introduction

- To improve monitoring of an active volcano we need to understand the multiple time scale components and variables that contribute to recorded measurements such as ground temperature.
- In this research, we examine how the atmospheric air and the meteorological conditions that take place on Santorini volcano monitoring station affect the surface volcanic temperature recordings of the volcano.
- The atmosphere significantly affects the surface's temperature of the volcano, and removal of the atmospheric effect is required for meaningful interpretations and reliable volcanic hazard evaluation.

Figure 1: A volcanic monitoring station on a fumarole which is located at an active volcano (Thera) at the Hellenic Volcanic Arc in South Aegean Sea, Greece.

Methods

- Time series of volcanic ground temperature from an active fumarole in Nea Kameni, Santorini were retrieved for the interval of June 2017 to May 2018. Meteorological data were acquired from the closest weather station.
- Time series analysis performed in SPSS statistical software at different time scales through a decomposition method.
- Detrending was performed with a 1st order differencing using 8-hours lag.
- The decomposition of the time series into seasonal-term and short-term components was performed by the application of the Kolmogorov-Zurbenko (KZ) filter to separate the noise that appears on the obtained data (Fig.2).
- Two mathematical models have been applied for the analysis, the Multiple Linear Regression (MLR) and the Time Series Regression (TSR) model.

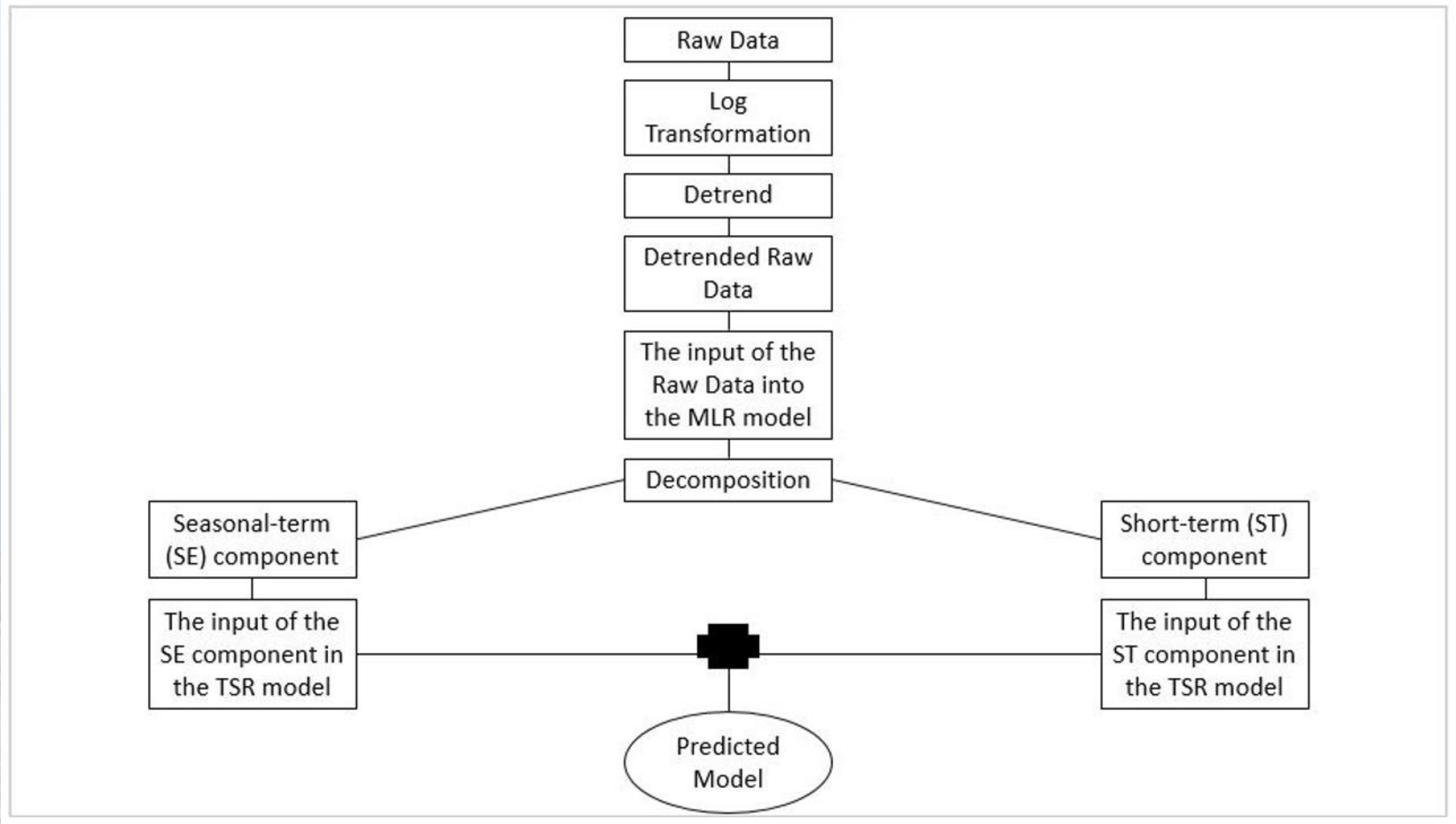

Figure 2: The flowchart of the methodology shows the integration of the two models at different stages of the overall data processing

Results

Table 1: The coefficient of determination (R^2) for seasonal and short-term component of the MLR and TSR models.

Component	R^2 of the MLR model	R^2 of the TSR model
Seasonal-term	0.853	0.961
Short-term	0.020	0.105

Table 2: The total explanation of the predicted TSR model for the fumarole's temperature.

Data Type	R^2	Variance %	Explanation	Total Explanation
Raw Data	0.186			
SE component	0.961	69.52	66.81	
ST component	0.105	30.48	3.20	
				70.01

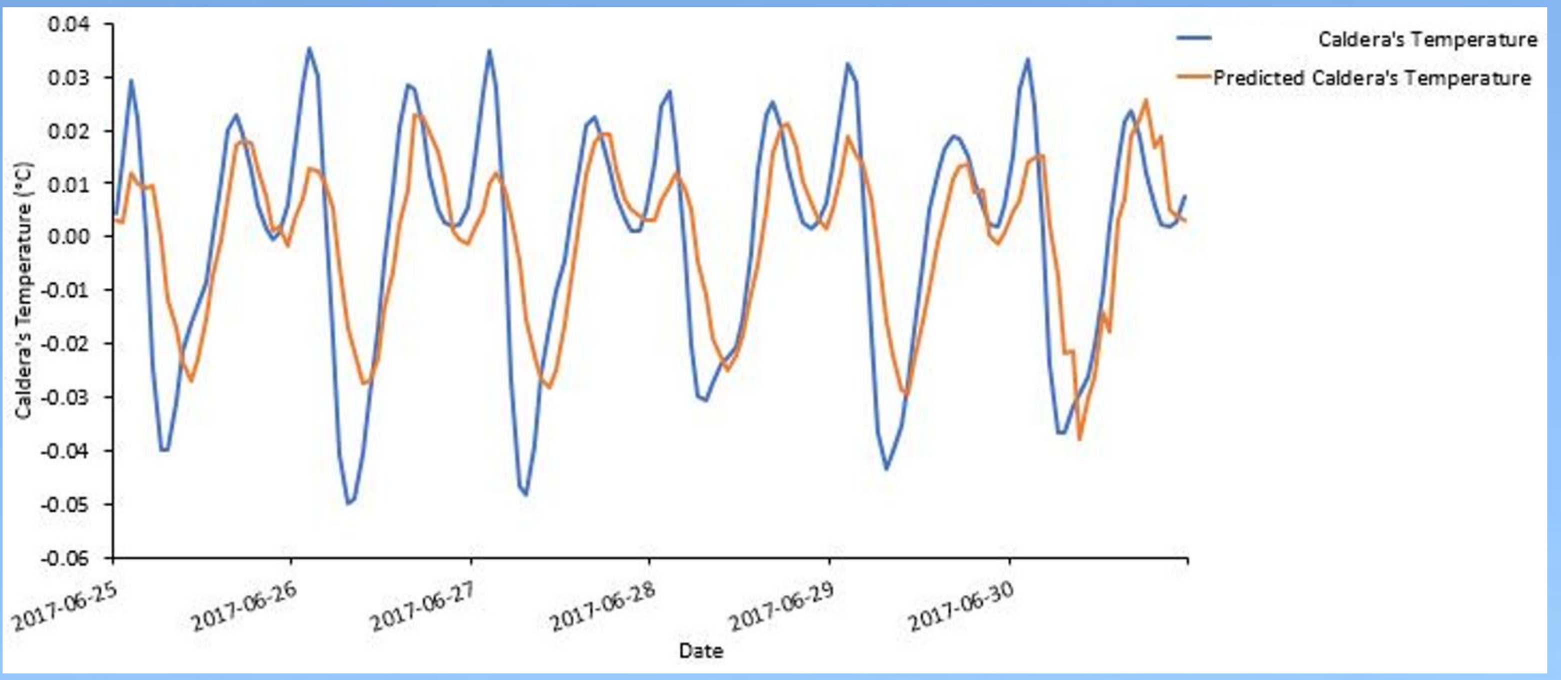

Figure 4: The predicted model against the seasonal-component of temperature data in Santorini fumarole.

Discussion

- Applying various degrees of differencing was unsuccessful to provide a stationary time series.
- A successful detrending using 8-hours lag of differencing may reveal a hidden periodicity of an 8-hours signal but this may require a further research.
- Unlike high temperature volcanoes, low temperature volcanoes are possibly more influenced by phenomena like atmosphere and monthly cycles, as supported by our study. However, other factors like sea level, 11-year cycles/ extraterrestrial patterns, tectonics, or other factors could also impact the volcano's behavior.
- Incorporating the TSR model increases the physical interpretation, and a more accurate prediction model is developed.
- A long-term component with a period of more than 12-months could not be determined due to the limitation of data records (12-months). A possible explanation may be that a larger periodicity of 365-days may occur in Santorini volcano.

Conclusion

- Removal of the noise with the application of decomposition at various time scales has facilitated an optimal prediction.
- In comparison to the MLR model's forecast, the TSR model has performed better by increasing the explanation (R^2) by up to seven times while it well interprets the effect of the atmospheric air on the surface caldera's temperature.
- Other processes besides atmospheric include but not limited to endogenic volcano-tectonic processes that are likely to affect the unexplained remnant part of the surface volcanic temperature of Santorini volcano.
- This methodology may be used in different volcanic regions to better comprehend the atmospheric contribution upon a volcano.

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